ON INTERIOR REGULARITY OF WEAK SOLUTIONS TO SECOND ORDER ELLIPTIC EQUATIONS

A PREPRINT

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ABSTRACT

For this project I will elaborate on the proofs of three theorems present in Evans *Partial Differential Equations* [1]. These are theorems describing the behavior of weak solutions to the second order elliptic equation $Lu = f$ on the interior of a bounded domain $\Omega \subset \mathbb{R}^n$.

1 Preliminaries

I will be addressing weak solutions to the boundary value problem [1] given by

$$\text{BVP} = \begin{cases} Lu = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{1}$$

where $\Omega \subset \mathbb{R}^n$ is open bounded. The operator has the form

$$Lu = -\sum_{i,j=1}^{n} \left(a^{ij}(x)u_{x_i}\right)_{x_j} + \sum_{i=1}^{n} b^i(x)u_{x_i} + c(x)u \tag{2}$$

or

$$Lu = -\sum_{i,j=1}^{n} a^{ij}(x)u_{x_i x_j} + \sum_{i=1}^{n} b^i(x)u_{x_i} + c(x)u \tag{3}$$

First, let us recall some basic definitions and theorems that will be utilized in the following proofs. Recall that the space $L^p(\Omega)$ is defined to be [2] the set of functions

$$L^p(\Omega) = \{f : \Omega \to \mathbb{R} \text{ measurable} |\ \|f\|_{L^p(\Omega)} < \infty\}.$$

Where the norm is defined to be

$$\|f\|_{L^p(\Omega)} = \left(\int_\Omega |f|^p\, dx\right)^{1/p}$$

I will also utilize the definition of the Sobolev space and the corresponding norm [1] given by

$$W^{k,p}(\Omega) = \{u \in L^p(\Omega) |\ u : \Omega \to \mathbb{R} \text{ such that } \forall \alpha \leq k,\ D^\alpha u \in L^p(\Omega) \text{ exists weakly}\}$$

where the norm in this space is given by

$$\|u\|_{W^{k,p}(\Omega)} = \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|^p_{L^p(\Omega)} \right)^{1/p} \approx \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(\Omega)}.$$

We also have the space defined $H^k(\Omega) = W^{k,2}(\Omega)$, which is a Hilbert space. Next I will prove the Cauchy Inequality on ϵ that I reference in some of my calculations:

Proof. Let $a, b \in \mathbb{R}$ and in particular $a, b > 0$ and let $\epsilon > 0$ then

$$\begin{aligned} ab &= ab \cdot \left(\frac{(2\epsilon)^{1/2}}{(2\epsilon)^{1/2}} \right) = \left(a(2\epsilon)^{1/2} \right) \left(\frac{b}{(2\epsilon)^{1/2}} \right) \\ &\leq \frac{\left(a(2\epsilon)^{1/2} \right)^2}{2} + \frac{\left(\frac{b}{(2\epsilon)^{1/2}} \right)^2}{2} \\ &= \frac{a^2 \cdot 2\epsilon}{2} + \frac{b^2 \cdot \frac{1}{2\epsilon}}{2} \\ &= a^2\epsilon + \frac{b^2}{4\epsilon}. \end{aligned}$$

□

We will also reference the definition of a *weak solution* of the boundary value problem (1):

Definition 1.1 (Weak Solution)**.**

(i) The bilinear form $B[\ ,\ ]$ associated with the divergence form elliptic operator L defined by (2) is

$$B[u, v] := \int_\Omega \sum_{i,j=1}^n a^{ij} u_{x_i} v_{x_j} + \sum_{i=1}^n b^i u_{x_i} v + cuv\ dx \tag{4}$$

for $u, v \in H_0^1(\Omega)$.

(ii) and we say that $u \in H_0^1(\Omega)$ is a weak solution of the BVP (1) if

$$B[u, v] = (f, v) := \int_\Omega fv\ dx$$

for all $v \in H_0^1(\Omega)$ where $(\ ,\ )$ denotes the inner product in $L^2(\Omega)$.

We will also utilize the difference quotient

$$D_k^\epsilon u(x) = \frac{u(x + \epsilon e_k) - u(x)}{\epsilon}$$

for $\epsilon \in \mathbb{R}$ and $|\epsilon| > 0$ very small, and note that e_k is the $k-$th standard orthonormal vector, i.e., $e_1 = \{1, 0, 0, \ldots, 0\}$. Here I will prove the following two formulas:

$$\int_\Omega v D_k^{-\epsilon} w\ dx = -\int_\Omega w D_k^\epsilon v dx \quad \text{for } w \in C_0^\infty(\Omega), \tag{5}$$

and

$$D_k^\epsilon(vw) = v^\epsilon D_k^\epsilon w + w D_k^\epsilon v \tag{6}$$

where $v^\epsilon := v(x + \epsilon e_k)$.

Proof. (5) Let $w \in C_0^\infty(\Omega)$, that is compact support in Ω and $\epsilon > 0$ very very very small. Now consider

$$\int_\Omega v(x)w(x-\epsilon e_k)\,dx = \int_{\Omega-\epsilon e_k} v(x+\epsilon e_k)w(x)\,dx = (\star),$$

however consider the figure below

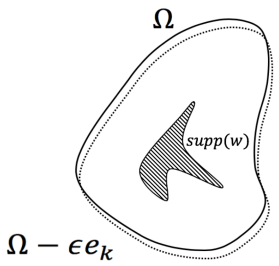


Figure 1: w has compact support in Ω, and $\Omega - \epsilon e_k$ is shifted by some small amount $\epsilon \neq 0$.

If ϵ is small, then we are effectively still containing the support of w and the preceding integral will hold over Ω instead. That is

$$(\star) = \int_\Omega v(x+\epsilon e_k)w(x)\,dx,$$

then subtract from both sides to get

$$\int_\Omega v(x)w(x-\epsilon e_k) - v(x)w(x)\,dx = \int_\Omega v(x+\epsilon e_k)w(x) - v(x)w(x)\,dx.$$

Divide by ϵ on both sides and reorganize terms to get

$$\int_\Omega vD_k^{-\epsilon}w\,dx = \int_\Omega v(x)\frac{w(x-\epsilon e_k)-w(x)}{\epsilon}\,dx = \int_\Omega w(x)\frac{v(x+\epsilon e_k)-v(x)}{\epsilon}\,dx = -\int_\Omega wD_k^\epsilon v\,dx$$

which is the desired result. □

Proof. (6) We have

$$\begin{aligned}
D_k^\epsilon(vw) &= \frac{v(x+\epsilon e_k)w(x+\epsilon e_k) - v(x)w(x)}{\epsilon} \\
&= \frac{v(x+\epsilon e_k)w(x+\epsilon e_k) - v(x+\epsilon e_k)w(x) + v(x+\epsilon e_k)w(x) - v(x)w(x)}{\epsilon} \\
&= v(x+\epsilon e_k)\frac{w(x+\epsilon e_k)-w(x)}{\epsilon} + w(x)\frac{v(x+\epsilon e_k)-v(x)}{\epsilon} \\
&= v^\epsilon D_k^\epsilon w + wD_k^\epsilon v
\end{aligned}$$

□

I will also be utilizing the definition of ellipticity [1] given below:

Definition 1.2. (Ellipticity) We say that the partial differential operator is (uniformly) elliptic if there exists a constant $\theta > 0$, positive, such that

$$\sum_{i,j=1}^n a^{ij}(x)\xi_i\xi_j \geq \theta|\xi|^2$$

for almost every $x \in \Omega$ and all $\xi \in \mathbb{R}^n$, and θ is called the ellipticity constant.

I will also reference the following theorem involving difference quotients and weak derivatives:

Theorem 1.1 (Difference Quotients and Weak Derivatives)**.**

(i) Suppose $1 \leq p < \infty$ and $u \in W^{1,p}(\Omega)$. Then for each $V \subset \Omega$ we have

$$\|D^\epsilon u\|_{L^p(V)} \leq C\|Du\|_{L^p(\Omega)}$$

for some constant C and all $0 < |h| < \frac{1}{2}\text{dist}(V, \partial\Omega)$.

(ii) Assume $1 < p < \infty$, $u \in L^p(V)$, and there exists a constant C such that $\|D^\epsilon u\|_{L^p(V)} \leq C$ for all $0 < |h| < \frac{1}{2}\text{dist}(V, \partial\Omega)$. Then $u \in W^{1,p}(V)$, with $\|Du\|_{L^p(V)} \leq C$.

Proof. First assume $1 \leq p < \infty$ and that u is a smooth function. Then for each $x \in V$, $i = 1, 1 \ldots, n$, and $0 < |\epsilon| < \frac{1}{2}\text{dist}(V, \partial\Omega)$ we observe

$$\frac{u(x + \epsilon e_k) - u(x)}{\epsilon} = \frac{u(x + 1 \cdot \epsilon e_k) - u(x + 0 \cdot \epsilon e_k)}{\epsilon}$$

But notice that

$$\int_0^1 u_{x_k}(x + t\epsilon e_k)\, dt = \frac{1}{\epsilon} u(x + t\epsilon e_k)\Big|_0^1 = \frac{u(x + 1 \cdot \epsilon e_k) - u(x + 0 \cdot \epsilon e_k)}{\epsilon};$$

therefore,

$$u(x + \epsilon e_k) - u(x) = \epsilon \int_0^1 u_{x_k}(x + t\epsilon e_k)\, dt.$$

Then

$$|u(x + \epsilon e_k) - u(x)| \leq |\epsilon| \int_0^1 |u_{x_k}(x + t\epsilon e_k)|\, dt \leq |\epsilon| \int_0^1 |Du(x + t\epsilon e_k)|\, dt,$$

this is equivalent to saying

$$|D_k^\epsilon u(x)| = \frac{|u(x + \epsilon e_k) - u(x)|}{|\epsilon|} \leq \int_0^1 |Du(x + t\epsilon e_k)|\, dt.$$

Then apply Holder's inequality to get

$$|D_k^\epsilon u(x)| \leq \left(\int_0^1 |Du(x + t\epsilon e_k)|^p\, dt \right)^{1/p} \implies |D_k^\epsilon u(x)|^p \leq \int_0^1 |Du(x + t\epsilon e_k)|^p\, dt$$

and then integrate over V to get

$$\int_V |D_k^\epsilon u(x)|^p\, dx \leq \int_V \int_0^1 |Du(x + t\epsilon e_k)|^p\, dt\, dx \leq C \sum_{i=1}^n \int_V \int_0^1 |Du(x + t\epsilon e_k)|^p\, dt\, dx$$

$$= C \sum_{i=1}^n \int_0^1 \int_V |Du(x + t\epsilon e_k)|^p\, dx\, dt.$$

We can interchange the integral symbols since $u \in W^{1,p}(\Omega)$; therefore $\|Du\|_{L^p(\Omega)} < \infty$ and then

$$\int_0^1 \int_V |Du(x + t\epsilon e_k)|^p\, dx\, dt < \infty,$$

this is an application of Fubini-Tonelli. Therefore, we arrive at

$$\int_V |D_k^\epsilon u(x)|^p\, dx \leq C \int_\Omega |Du|^p\, dx.$$

Now suppose that $\|D^\epsilon u\|_{L^p(V)} \leq C$ for $0 < |\epsilon| < \text{dist}(V, \partial\Omega)$, and let $i \in [1, \ldots, n]$ and $\phi \in C_0^\infty(V)$, that is a smooth function with compact support in V. Recall that for small $\epsilon > 0$ we can utilize (5), the integration by parts formula for difference quotients:

$$\int_V u(D_k^\epsilon \phi)\, dx = -\int_V (D_k^{-\epsilon} u)\phi\, dx$$

Further by our assumption we have that $D_k^{-\epsilon} u$ is uniformly bounded that is

$$\sup_\epsilon \|D_k^{-\epsilon} u\|_{L^p(V)} < \infty;$$

therefore, if we couple this with $1 < p < \infty$, then there exists a function $w_k \in L^p(V)$ such that $D_k^{-\epsilon_j} u \to w_k$ weakly in $L^p(V)$ for a subsequence $\epsilon_j \to 0$. Therefore

$$\begin{aligned}\int_V u\phi_{x_k}\, dx &= \int_\Omega u\phi_{x_k}\, dx = \int_\Omega u \lim_{\epsilon_j \to 0} D_k^{\epsilon_j}\phi\, dx = \lim_{\epsilon_j \to 0} \int_\Omega u D_k^{\epsilon_j}\phi\, dx \\ &= -\lim_{\epsilon_j \to 0} \int_\Omega D_k^{-\epsilon_j} u\phi\, dx = -\int_\Omega w_k \phi\, dx.\end{aligned}$$

Thus $u_{x_k} = w_k$ weakly for $k = 1, \ldots, n$; therefore, $Du \in L^p(V)$. Since by hypothesis assumption $u \in L^p(V)$, then $u \in W^{1,p}(V)$. □

Now I will state the General Sobolev Inequalities from [1].

Theorem 1.2 (General Sobolev inequalities)**.** Let Ω be a bounded open subset of $\mathbb{R}^n$, with C^1 boundary. Assume $u \in W^{k,p}(\Omega)$.

(i) If $k < n/p$ then $u \in L^q(\Omega)$ where

$$\frac{1}{q} = \frac{1}{p} - \frac{k}{n}.$$

With the estimate

$$\|u\|_{L^q(\Omega)} \leq C\|u\|_{W^{k,p}(\Omega)}$$

and $C = C(k, p, n, \Omega)$.

(ii) If $k > n/p$, then $u \in C^{k-[n/p]-1,\gamma}(\overline{\Omega})$ where

$$\gamma = \begin{cases} \left[\frac{n}{p}\right] + 1 - \frac{n}{p}, & \frac{n}{p} \notin \mathbb{Z} \\ \text{any positive number} < 1, & \frac{n}{p} \in \mathbb{Z} \end{cases}$$

with the estimate

$$\|u\|_{C^{k-[n/p]-1,\gamma}(\overline{\Omega})} \leq C\|u\|_{W^{k,p}(\Omega)}$$

where $C = C(k, p, n, \gamma, \Omega)$.

2 Interior H^2-regularity

For this theorem we are concerned with the interior of a region $\Omega \subset \mathbb{R}^n$ and therefore we do not need $u \in H_0^1(\Omega)$ (and we don't necessarily require that $u = 0$ on $\partial\Omega$. I will state the theorem from [1] below:

Theorem 2.1 (Interior H^2-regularity)**.** Assume that $a^{ij} \in C^1(\Omega)$, $b^i, c \in L^\infty(\Omega)$ for $i, j = 1, \ldots, n$ and $f \in L^2(\Omega)$. Further suppose that $u \in H^1(\Omega)$ is a weak solution of the elliptic partial differential equation $Lu = f$ in Ω. Then $u \in H^2_{\text{loc}}(\Omega)$; and for each open subset $V \Subset \Omega$ we have the estimate

$$\|u\|_{H^2(V)} \leq C \left(\|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)} \right) \tag{7}$$

where $C = C(V, \Omega, a^{ij}, b^i, c)$.

Now I will begin the proof following Evans, however I will be filling the "gaps" between lines of his proof and be more rigorous.

Proof. The first three parts of this proof are setting up the components we will be estimating. Parts 4 and 5 are where we will be estimating components, and finally we will incorporate the separate estimates into one final estimate.

Part 1. First, we will fix $V \Subset \Omega \subset \mathbb{R}^n$ which is bounded and open and we will choose a set W such that $V \Subset W \Subset \Omega$, below is a nice representation

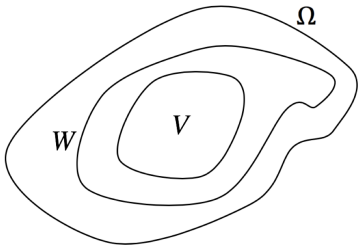


Figure 2: The regions $V \Subset W \Subset \Omega$

Now we will select a smooth function $\zeta(x) \in C^\infty(\mathbb{R}^n)$ such that

$$\zeta(x) = \begin{cases} 1 & \text{on } V, \\ 0 & \text{on } \mathbb{R}^n \setminus W, \\ [0,1] & \text{on } W \setminus V. \end{cases}$$

This is called a cutoff function, we utilize this to restrict our functions and expression to region contain entirely with in Ω and not including the boundary, since our intent is to examine the *interior* behavior of our solution u and the hypothesis of our theorem does not include information regarding the boundary $\partial\Omega$.

Part 2. By assumption of the theorem, that is u is a weak solution of (1), then by Definition 1.1 we have that $B[u, v] = \int_\Omega fv \, dx$ for all $v \in H^1_0(\Omega)$ and the left hand side $B[u, v]$ is given by (4). Therefore we have the following equation

$$B[u, v] = \int_\Omega \sum_{i,j=1}^n a^{ij} u_{x_i} v_{x_j} + \sum_{i=1}^n b^i u_{x_i} v + cuv \, dx = \int_\Omega fv \, dx = (f, v), \tag{8}$$

then separate the integrals on the left hand side of (8) to get

$$\int_\Omega \sum_{i,j=1}^n a^{ij} u_{x_i} v_{x_j} \, dx + \int_\Omega \sum_{i=1}^n b^i u_{x_i} v + cuv \, dx = \int_\Omega fv \, dx,$$

and subtract the second integral on the left hand side to the right hand side. That is

$$\int_\Omega \sum_{i,j=1}^n a^{ij} u_{x_i} v_{x_j}\, dx = \int_\Omega fv\, dx - \int_\Omega \sum_{i=1}^n b^i u_{x_i} v - cuv\, dx = \int_\Omega \left(f - \sum_{i=1}^n b^i u_{x_i} - cu \right) v\, dx = \int_\Omega \tilde{f} v\, dx,$$

or simply put

$$\sum_{i,j=1}^n \int_\Omega a^{ij} u_{x_i} v_{x_j}\, dx = \int_\Omega \tilde{f} v\, dx, \tag{9}$$

we can move the summation outside the integrand since it is finite. And we define

$$\tilde{f} := f - \sum_{i=1}^n b^i u_{x_i} - cu. \tag{10}$$

Part 3. Next we will set $|\epsilon| > 0$ small and let $k \in \{1, \dots, n\}$ and then define the function

$$v := -D_k^{-\epsilon}(\zeta^2 D_k^\epsilon u) \tag{11}$$

and substitute v into (9) to get

$$A := -\sum_{i,j=1}^n \int_\Omega a^{ij} u_{x_i} \left(D_k^{-\epsilon}(\zeta^2 D_k^\epsilon u) \right)_{x_j}\, dx = -\int_\Omega \tilde{f} D_k^{-\epsilon}(\zeta^2 D_k^\epsilon u)\, dx =: B \tag{12}$$

Part 4. Let us now estimate A, first observe in general that $(D_k^\epsilon f)_{x_j} = D_k^\epsilon (f_{x_j})$; therefore we get

$$A = -\sum_{i,j=1}^n \int_\Omega a^{ij} u_{x_i} \left(D_k^{-\epsilon}(\zeta^2 D_k^\epsilon u) \right)_{x_j}\, dx = -\sum_{i,j=1}^n \int_\Omega a^{ij} u_{x_i} D_k^{-\epsilon}((\zeta^2 D_k^\epsilon u)_{x_j})\, dx.$$

Now apply (5) where $v = a^{ij} u_{x_i}$ and $w = (\zeta^2 D_k^\epsilon u)_{x_j}$ to get

$$A = \sum_{i,j=1}^n \int_\Omega D_k^\epsilon (a^{ij} u_{x_i})(\zeta^2 D_k^\epsilon u)_{x_j}\, dx.$$

Next apply (6) to $D_k^\epsilon (a^{ij} u_{x_i})$ and you get

$$A = \sum_{i,j=1}^n \int_\Omega a^{ij,\epsilon} D_k^\epsilon u_{x_i} (\zeta^2 D_k^\epsilon u)_{x_j} + u_{x_i} D_k^\epsilon a^{ij} (\zeta^2 D_k^\epsilon u)_{x_j}\, dx. \tag{13}$$

Now let us expand the integrand, by noting

$$(\zeta^2 D_k^\epsilon u)_{x_j} = 2\zeta \zeta_{x_j} D_k^\epsilon u + \zeta^2 D_k^\epsilon u_{x_j}$$

and substitute this into (13) to get

$$A = \sum_{i,j=1}^n \int_\Omega 2\zeta\zeta_{x_j} a^{ij,\epsilon} D_k^\epsilon u_{x_i} D_k^\epsilon u + \zeta^2 a^{ij,\epsilon} D_k^\epsilon u_{x_i} D_k^\epsilon u_{x_j} + 2\zeta\zeta_{x_j} u_{x_i} D_k^\epsilon a^{ij} D_k^\epsilon u + \zeta^2 u_{x_i} D_k^\epsilon a^{ij} D_k^\epsilon u_{x_j}\, dx.$$

We separate the above expression into two parts, namely

$$A_1 := \sum_{i,j=1}^n \int_\Omega \zeta^2 a^{ij,\epsilon} D_k^\epsilon u_{x_i} D_k^\epsilon u_{x_j}\, dx \tag{14}$$

$$A_2 := \sum_{i,j=1}^n \int_\Omega 2\zeta\zeta_{x_j} a^{ij,\epsilon} D_k^\epsilon u_{x_i} D_k^\epsilon u + 2\zeta\zeta_{x_j} u_{x_i} D_k^\epsilon a^{ij} D_k^\epsilon u + \zeta^2 u_{x_i} D_k^\epsilon a^{ij} D_k^\epsilon u_{x_j}\, dx \tag{15}$$

Carefully notice in (14) we quickly apply ellipticity Definition 1.2 to get

$$A_1 = \sum_{i,j=1}^{n} \int_\Omega \zeta^2 a^{ij,\epsilon} D_k^\epsilon u_{x_i} D_k^\epsilon u_{x_j}\, dx = \int_\Omega \zeta^2 \sum_{i,j=1}^{n} a^{ij,\epsilon} D_k^\epsilon u_{x_i} D_k^\epsilon u_{x_j}\, dx \geq \int_\Omega \zeta^2 \theta |D_k^\epsilon Du|^2\, dx = \theta \|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)}. \tag{16}$$

For (15) we will take the absolute value and examine each summation. Recall that $a^{ij} \in C^1(\Omega)$ and ζ is smooth with compact support; therefore, a^{ij}, ζ_{x_i} are bounded. We have

$$S_1 = \int_\Omega 2\zeta \sum_{i,j=1}^{n} \zeta_{x_j} a^{ij,\epsilon} D_k^\epsilon u_{x_i} D_k^\epsilon u\, dx \Longrightarrow |S_1| \leq \int_\Omega 2\zeta \sum_{i,j=1}^{n} |\zeta_{x_j}||a^{ij,\epsilon}||D_k^\epsilon u_{x_i}||D_k^\epsilon u|\, dx \leq C_1 \int_\Omega \zeta |D_k^\epsilon Du||D_k^\epsilon u|\, dx.$$

Next we have

$$S_2 = \int_\Omega 2\zeta \sum_{i,j=1}^{n} \zeta_{x_j} u_{x_i} D_k^\epsilon a^{ij} D_k^\epsilon u\, dx \Longrightarrow |S_2| \leq \int_\Omega 2\zeta \sum_{i,j=1}^{n} |\zeta_{x_j}||D_k^\epsilon a^{ij}||u_{x_i}||D_k^\epsilon u|\, dx \leq C_2 \int_\Omega \zeta |D_k^\epsilon u||Du|\, dx.$$

And next we have

$$S_3 = \int_\Omega \zeta^2 \sum_{i,j=1}^{n} u_{x_i} D_k^\epsilon a^{ij} D_k^\epsilon u_{x_j}\, dx \Longrightarrow |S_3| \leq \int_\Omega \zeta \sum_{i,j=1}^{n} |\zeta||D_k^\epsilon a^{ij}||u_{x_i}||D_k^\epsilon u_{x_j}|\, dx \leq C_3 \int_\Omega \zeta |D_k^\epsilon Du||Du|\, dx.$$

Combine these to get an estimate for (15), that is

$$|A_2| \leq |S_1| + |S_2| + |S_3| \leq C \int_\Omega \zeta |D_k^\epsilon Du||D_k^\epsilon u| + \zeta |D_k^\epsilon u||Du| + \zeta |D_k^\epsilon Du||Du|\, dx \tag{17}$$

In (17) we need to separate out the multiplied terms; therefore, we will utilize Cauchy's inequality for a given $\delta > 0$ (since all the terms are positive). Recall that this inequality is given by

$$ab \leq \delta a^2 + \frac{b^2}{4}\delta$$

where $a, b > 0$ and $\delta > 0$. Therefore, let us look at each term in (17), we have

$$(i)\ \ \zeta |D_k^\epsilon Du||D_k^\epsilon u| \leq \delta \zeta^2 |D_k^\epsilon Du|^2 + \frac{|D_k^\epsilon u|^2}{4\delta}$$

$$(ii)\ \ \zeta |D_k^\epsilon u||Du| \leq \delta \zeta^2 |D_k^\epsilon u|^2 + \frac{|Du|^2}{4\delta}$$

$$(iii)\ \ \zeta |D_k^\epsilon Du||Du| \leq \delta \zeta^2 |D_k^\epsilon Du|^2 + \frac{|Du|^2}{4\delta}$$

we add up $(i) - (iii)$ and collect like terms to get

$$(i) - (iii) = 2\delta\zeta^2 |D_k^\epsilon Du|^2 + \left(\delta\zeta^2 + \frac{1}{4\delta}\right)|D_k^\epsilon u|^2 + \frac{1}{2\delta}|Du|^2 \leq \delta' \zeta^2 |D_k^\epsilon Du|^2 + \frac{C_6 \zeta^2}{\delta'}\left(|D_k^\epsilon u|^2 + |Du|^2\right)$$

for $\delta' > 0$. Substitute this back into (17) to get

$$\begin{aligned} |A_2| &\leq C \int_\Omega \delta' \zeta^2 |D_k^\epsilon Du|^2 + \frac{C_6 \zeta^2}{\delta'}\left(|D_k^\epsilon u|^2 + |Du|^2\right)\, dx \\ &\leq \rho \int_\Omega \zeta^2 |D_k^\epsilon Du|^2\, dx + \frac{\tilde{C}}{\rho} \int_\Omega \zeta^2 \left(|D_k^\epsilon u|^2 + |Du|^2\right)\, dx \end{aligned}$$

we will utilize the compact support of ζ on the second integral to get

$$\begin{aligned} |A_2| &\leq \rho \int_\Omega \zeta^2 |D_k^\epsilon Du|^2\, dx + \frac{\tilde{C}}{\rho} \int_W |D_k^\epsilon u|^2 + |Du|^2\, dx \\ &= \rho \|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} + \frac{\tilde{C}}{\rho}\left(\|D_k^\epsilon u\|^2_{L^2(W)} + \|Du\|^2_{L^2(W)}\right). \end{aligned} \tag{18}$$

Now recall Theorem 1.1(i) since $W \Subset \Omega$ and we have that $\|D_k^\epsilon u\|^2_{L^2(W)} \leq C\|Du\|_{L^2(\Omega)}$. If we use this in (18) we get that

$$\begin{aligned}|A_2| &\leq \rho\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} + \frac{\tilde{C}}{\rho}\left(\|Du\|^2_{L^2(\Omega)} + \|Du\|^2_{L^2(W)}\right)\\ &\leq \rho\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} + \frac{\tilde{C}}{\rho}\left(2\|Du\|^2_{L^2(\Omega)}\right)\\ &\leq \rho\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} + C\|Du\|^2_{L^2(\Omega)}.\end{aligned}$$

Since this holds for all $\rho > 0$, set $\rho = \theta/2$ and we get

$$|A_2| \leq \frac{\theta}{2}\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} + C\|Du\|^2_{L^2(\Omega)}. \tag{19}$$

This implies

$$-\frac{\theta}{2}\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} - C\|Du\|^2_{L^2(\Omega)} \leq A_2 \leq \frac{\theta}{2}\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} + C\|Du\|^2_{L^2(\Omega)},$$

add A_1 through out to get

$$-\frac{\theta}{2}\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} - C\|Du\|^2_{L^2(\Omega)} + A_1 \leq A \leq \frac{\theta}{2}\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} + C\|Du\|^2_{L^2(\Omega)} + A_1.$$

We will examine the left half inequality and use (16) to get that

$$\frac{\theta}{2}\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} - C\|Du\|^2_{L^2(\Omega)} \leq A, \tag{20}$$

this estimates A from (12).

Part 5. Let us estimate B from (12), recall (10) and (11) and that $B := \int_\Omega \tilde{f}v\,dx$. Also recall the hypothesis that $b^i, c \in L^\infty(\Omega)$ and notice that

$$|\tilde{f}| \leq |f| + \sum_{i=1}^n |b^i||u_{x_i}| + |c||u| \leq C\left(|f| + |Du| + |u|\right).$$

Therefore,

$$|B| \leq C\int_\Omega \left(|f| + |Du| + |u|\right)|v|\,dx \tag{21}$$

Now consider the following and apply Theorem 1.1(i), we have

$$\begin{aligned}\int_\Omega |v|^2\,dx &= \int_\Omega |D_k^{-\epsilon}(\zeta^2 D_k^\epsilon u)|^2\,dx \leq C\int_\Omega |D(\zeta^2 D_k^\epsilon u)|^2\,dx\\ &= C\int_\Omega |2\zeta D\zeta D_k^\epsilon u + \zeta^2 D_k^\epsilon Du|^2\,dx \qquad (22)\\ &= C\int_W |2\zeta D\zeta D_k^\epsilon u + \zeta^2 D_k^\epsilon Du|^2\,dx\\ \text{I applied Minkowski}\quad &= C\|2\zeta D\zeta D_k^\epsilon u + \zeta^2 D_k^\epsilon Du\|^2_{L^2(W)} \qquad (23)\\ &\leq C\left(\|2\zeta D\zeta D_k^\epsilon u\|_{L^2(W)} + \|\zeta^2 D_k^\epsilon Du\|_{L^2(W)}\right)^2\\ &\leq C\left(\|D_k^\epsilon u\|_{L^2(W)} + \|\zeta^2 D_k^\epsilon Du\|_{L^2(W)}\right)^2\\ &\leq C\left(\|D_k^\epsilon u\|^2_{L^2(W)} + \|\zeta^2 D_k^\epsilon Du\|^2_{L^2(W)}\right) \qquad (24)\\ &= C\int_W |D_k^\epsilon u|^2 + \zeta^2|D_k^\epsilon Du|^2\,dx\\ \text{I applied Thm. 1.1 here}\quad &\leq C\int_\Omega |Du|^2 + \zeta^2|D_k^\epsilon Du|^2\,dx. \qquad (25)\end{aligned}$$

Now with the integrand of (21) apply the Cauchy inequality again to get

$$\begin{aligned}|v|(|f|+|Du|+|u|) &= |v||f|+|v||Du|+|v||u| \\ &\leq \delta|v|^2+\frac{|f|^2}{4\delta}+\delta|v|^2+\frac{|Du|^2}{4\delta}+\delta|v|^2+\frac{|u|^2}{4\delta} \\ &\leq \delta'|v|^2+\frac{C}{\delta'}\left(f^2+u^2+|Du|^2\right).\end{aligned}$$

Then with (22) we have that

$$|B| \leq \delta' \int_\Omega \zeta^2 |D_k^\epsilon Du|^2 \, dx + \frac{C}{\delta'} \int_\Omega f^2+u^2+|Du|^2 \, dx. \tag{26}$$

Let $\delta' = \theta/4$ to get

$$|B| \leq \frac{\theta}{4} \int_\Omega \zeta^2 |D_k^\epsilon Du|^2 \, dx + C \int_\Omega f^2+u^2+|Du|^2 \, dx = \frac{\theta}{4}\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} + C \int_\Omega f^2+u^2+|Du|^2 \, dx. \tag{27}$$

Part 6. We now have all our estimates, now we utilize (20), (27), and (12). We have

$$A = B \Longrightarrow \frac{\theta}{2}\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} - C\|Du\|^2_{L^2(\Omega)} \leq A = B \leq \frac{\theta}{4}\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} + C \int_\Omega f^2+u^2+|Du|^2 \, dx.$$

Add and subtract corresponding terms to get

$$\frac{\theta}{4}\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} \leq C \int_\Omega f^2+u^2+|Du|^2 \, dx,$$

or simply combine constant to get

$$\|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} \leq C \int_\Omega f^2+u^2+|Du|^2 \, dx. \tag{28}$$

Since $V \Subset \Omega$ we can majorize the L^2 norm on the left-hand side to get

$$\|D_k^\epsilon Du\|^2_{L^2(V)} \leq \|\zeta D_k^\epsilon Du\|^2_{L^2(\Omega)} \leq C \int_\Omega f^2+u^2+|Du|^2 \, dx$$

for all $k = 1, \ldots, n$ and $\epsilon > 0$ very small. Since by hypothesis $u \in H^1(\Omega)$, then $u \in W^{1,2}(\Omega)$; therefore, from the last inequality above we have $\|D_k^\epsilon Du\|^2_{L^2(V)} < \infty$ for all $k = 1, \ldots, n$ and $\epsilon > 0$ very small. Therefore, if we let $\epsilon \to 0$ for every k component we get that $\|D^2 u\|_{L^2(V)} < \infty$ (note: we can use Lebesgue Dominated Convergence to pass the limit inside). That is

$$\lim_{\epsilon \to 0} \int_V |D_k^\epsilon Du|^2 \, dx = \int_V \lim_{\epsilon \to 0} |D_k^\epsilon Du|^2 \, dx = \int_V |D(Du)|^2 \, dx = \int_V |D^2 u|^2 \, dx = \|D^2 u\|^2_{L^2(V)}.$$

Thus $Du \in H^1_{loc}(\Omega \subset \mathbb{R}^n)$, locally because V is a compact set in Ω. Further observe, that since $u \in H^1(\Omega)$ then $u \in H^1(V)$ for $V \Subset \Omega$ and then $u \in H^1_{loc}(\Omega \subset \mathbb{R}^n)$. Therefore, we have that $u, Du, D^2 u \in L^2(V)$; thus $u \in H^2_{loc}(\Omega)$. Now let us examine

$$\begin{aligned}\|u\|_{H^2(V)} = \|u\|_{W^{2,2}(V)} &\leq C\left(\|u\|_{L^2(\Omega)} + \|Du\|_{L^2(\Omega)} + \|D^2 u\|_{L^2(\Omega)}\right) \\ &\leq C\left(\|u\|_{L^2(\Omega)} + \|Du\|_{L^2(\Omega)} + \|D_k^\epsilon Du\|_{L^2(\Omega)}\right) \\ &\leq C\left(\|u\|_{L^2(\Omega)} + \|Du\|_{L^2(\Omega)} + \left(C\int_\Omega f^2+u^2+|Du|^2 \, dx\right)^{1/2}\right) \\ &\leq C\left(C_1\|u\|_{L^2(\Omega)} + C_2\|Du\|_{L^2(\Omega)} + C_3\|f\|_{L^2(\Omega)}\right) \qquad (29) \\ &\leq C\left(\|u\|_{L^2(\Omega)} + \|Du\|_{L^2(\Omega)} + \|f\|_{L^2(\Omega)}\right) \\ &\leq C\left(\|u\|_{W^{1,2}(\Omega)} + \|f\|_{L^2(\Omega)}\right) = C\left(\|u\|_{H^1(\Omega)} + \|f\|_{L^2(\Omega)}\right) \qquad (30)\end{aligned}$$

Part 7. Next observe that since $V \Subset W \Subset \Omega$ then we can apply the same inequality above to get

$$\|u\|_{H^2(V)} \leq C \left(\|u\|_{H^1(W)} + \|f\|_{L^2(W)} \right). \tag{31}$$

Now select a different cutoff function such that

$$\zeta(x) = \begin{cases} 1 & \text{on } W, \text{ spt } \zeta \subset \Omega \\ 0 \leq \zeta \leq 1 \end{cases}$$

and let $v = \zeta^2 u$ and substitute this back into (9) to get

$$\sum_{i,j=1}^{n} \int_\Omega a^{ij} u_{x_i} (\zeta^2 u)_{x_j} \, dx = \int_\Omega \tilde{f} \zeta^2 u \, dx.$$

Then

$$\sum_{i,j=1}^{n} \int_\Omega \zeta^2 a^{ij} u_{x_i} u_{x_j} + 2\zeta \zeta_{x_j} a^{ij} u_{x_i} \, dx = \int_\Omega \zeta^2 f u - \zeta^2 \sum_{i=1}^{n} b^i u_{x_i} u - \zeta^2 c u^2 \, dx.$$

Collect terms to one side

$$\int_\Omega \sum_{i,j=1}^{n} \zeta^2 a^{ij} u_{x_i} u_{x_j} \, dx = \int_\Omega \zeta^2 f u - \zeta^2 \sum_{i=1}^{n} b^i u_{x_i} u - \sum_{i,j=1}^{n} 2\zeta \zeta_{x_j} a^{ij} u_{x_i} - \zeta^2 c u^2 \, dx.$$

We can apply ellipticity on the left hand side to get

$$\begin{aligned}
\int_\Omega \theta \zeta^2 |Du|^2 \, dx &\leq \int_\Omega \zeta^2 f u - \zeta^2 \sum_{i=1}^{n} b^i u_{x_i} u - \sum_{i,j=1}^{n} 2\zeta \zeta_{x_j} a^{ij} u_{x_i} - \zeta^2 c u^2 \, dx \\
&\leq \int_\Omega \zeta^2 f u - \zeta^2 c u^2 \, dx \leq \int_\Omega \zeta^2 |f||u| + \zeta^2 |c| u^2 \, dx \\
\text{I applied Cauchy inequality here } &\leq \int_\Omega \zeta^2 \delta f^2 + \zeta^2 \frac{u^2}{4\delta} + \zeta^2 |c| u^2 \, dx \\
&\leq \int_\Omega \delta f^2 + \left(|c| + \frac{1}{4\delta} \right) u^2 \, dx \\
&\leq C \int_\Omega f^2 + u^2 \, dx = C \left(\|f\|^2_{L^2(\Omega)} + \|u\|^2_{L^2(\Omega)} \right).
\end{aligned}$$

Therefore,

$$\|Du\|_{L^2(\Omega)} \leq C \left(\|f\|^2_{L^2(\Omega)} + \|u\|^2_{L^2(\Omega)} \right),$$

for all $\theta > 0$. Now observe

$$\|u\|_{H^1(W)} \leq \|u\|_{L^2(W)} + \|Du\|_{L^2(W)} \leq \|u\|_{L^2(\Omega)} + \|Du\|_{L^2(\Omega)} \leq C \left(\|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)} \right).$$

Couple this with (31) to get

$$\|u\|_{H^2(V)} \leq C \left(\|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)} \right).$$

□

3 Higher Interior Regularity

Theorem 3.1 (Higher interior regularity)**.** Let m be a non-negative integer, and assume

$$a^{ij}, b^i, c \in C^{m+1}(\Omega) \tag{32}$$

for $i, j = 1, \ldots, n$ and

$$f \in H^m(\Omega). \tag{33}$$

Suppose $u \in H^1(\Omega)$ is a weak solution of the elliptic PDE, $Lu = f$ in Ω. Then

$$u \in H^{m+2}_{loc}(\Omega); \tag{34}$$

and for each $V \Subset \Omega$ we have the estimate

$$\|u\|_{H^{m+2}(V)} \leq C \left(\|f\|_{H^m(\Omega)} + \|u\|_{L^2(\Omega)} \right), \tag{35}$$

where the constant $C = C(m, \Omega, V, a^{ij}, b^i, c)$.

Proof. To begin this proof we will start with induction on (34) and (35).

Part 1. Therefore assume that (34) and (35) hold for a non-negative integer m and all open sets Ω, coefficients $a^{ij}, b^i, c \in C^{m+1}(\Omega)$, and $f \in H^m(\Omega), u$ as stated in the hypothesis. Now we begin induction, suppose

$$a^{ij}, b^i, c \in C^{(m+1)+1}(\Omega) = C^{m+2}(\Omega) \tag{36}$$

and

$$f \in H^{m+1}(\Omega) \tag{37}$$

and by problem hypothesis $u \in H^1(\Omega)$ is a weak solution of $Lu = f$ in Ω, then by induction hypothesis $u \in H^{m+2}_{loc}$ we have the estimate

$$\|u\|_{H^{m+2}(W)} \leq C \left(\|f\|_{H^m(\Omega)} + \|u\|_{L^2(\Omega)} \right), \tag{38}$$

for each $W \Subset \Omega$ and $C(W, a^{ij}, b^i, c)$. Now fix $V \Subset W \Subset \Omega$, refer to Figure 2.

Part 2. Now consider α any multiindex with

$$|\alpha| = m + 1, \tag{39}$$

and let $\phi \in C_0^\infty(\Omega)$ be an arbitrary test function, that is a smooth function with compact support spt $\phi \subset \Omega$. Define

$$v := (-1)^{|\alpha|} D^\alpha \phi$$

and substitute v into $B[u, v] = (f, v)_{L^2(\Omega)}$, that is (8) to get $B[u, v] =$

$$\int_\Omega \sum_{i,j=1}^n a^{ij} u_{x_i} \left((-1)^{|\alpha|} D^\alpha \phi \right)_{x_j} + \sum_{i=1}^n b^i u_{x_i} \left((-1)^{|\alpha|} D^\alpha \phi \right) + cu \left((-1)^{|\alpha|} D^\alpha \phi \right) \, dx = \int_\Omega f (-1)^{|\alpha|} D^\alpha \phi \, dx. \tag{40}$$

For the right hand side of (40) we can use the definition of weak partial derivative to get

$$\int_\Omega f (-1)^{|\alpha|} D^\alpha \phi \, dx = \int_\Omega \phi D^\alpha f \, dx \tag{41}$$

Now Evans claims that we need to *perform some integration by parts* to get the next identity, but this is not trivial to do, I will break down the left hand side of (40) into three parts.

(i) Let us examine the integral

$$\begin{aligned}\int_\Omega a^{ij}u_{x_i}\left((-1)^{|\alpha|}D^\alpha\phi\right)_{x_j}\,dx &= (-1)^{|\alpha|}\int_\Omega a^{ij}u_{x_i}\left(D^\alpha\phi\right)_{x_j}\,dx = (-1)^{|\alpha|}\int_\Omega a^{ij}u_{x_i}D^\alpha\phi_{x_j}\,dx\\ &= \int_\Omega \phi_{x_j}D^\alpha(u_{x_i}a^{ij})\,dx\\ \text{Apply Leibniz Rule} &= \int_\Omega \phi_{x_j}\sum_{\beta\le\alpha}\binom{\alpha}{\beta}D^\beta u_{x_i}D^{\alpha-\beta}a^{ij}\,dx\\ \text{Integration by parts } &= -\int_\Omega \phi\sum_{\beta\le\alpha}\binom{\alpha}{\beta}\left(D^\beta u_{x_i}D^{\alpha-\beta}a^{ij}\right)_{x_j}\,dx \end{aligned} \tag{42}$$

(ii) Let us examine the integral

$$\begin{aligned}\int_\Omega b^i u_{x_i}\left((-1)^{|\alpha|}D^\alpha\phi\right)\,dx &= (-1)^{|\alpha|}\int_\Omega b^i u_{x_i}D^\alpha\phi\,dx = \int_\Omega \phi D^\alpha(b^i u_{x_i})\,dx\\ &= \int_\Omega \phi\sum_{\beta\le\alpha}\binom{\alpha}{\beta}D^\beta u_{x_i}D^{\alpha-\beta}b^i\,dx. \end{aligned} \tag{43}$$

(iii) Let us examine the integral

$$\begin{aligned}\int_\Omega cu\left((-1)^{|\alpha|}D^\alpha\phi\right)\,dx &= (-1)^{|\alpha|}\int_\Omega cuD^\alpha\phi\,dx = \int_\Omega \phi D^\alpha(cu)\,dx\\ &= \int_\Omega \phi\sum_{\beta\le\alpha}\binom{\alpha}{\beta}D^\beta uD^{\alpha-\beta}c\,dx. \end{aligned} \tag{44}$$

Now put (42), (43), and (44) back into the left hand side of (40) to get

$$\int_\Omega \phi\sum_{\beta\le\alpha}\binom{\alpha}{\beta}\underbrace{\left(-\sum_{i,j=1}^n\left(D^\beta u_{x_i}D^{\alpha-\beta}a^{ij}\right)_{x_j}+\sum_{i=1}^n D^\beta u_{x_i}D^{\alpha-\beta}b^i+D^\beta uD^{\alpha-\beta}c\right)}_{\text{stuff}}\,dx = \int_\Omega \phi D^\alpha f\,dx$$

To avoid re-writing the large parenthetical expression over again, I will refer to it as "stuff". I will split the outer summation $\beta\le\alpha$ in the following way

$$\int_\Omega \phi\sum_{\beta\le\alpha}\binom{\alpha}{\beta}(\text{stuff})\,dx = \underbrace{\int_\Omega \phi\sum_{\beta=\alpha}\binom{\alpha}{\beta}(\text{stuff})\,dx}_{I_1}+\underbrace{\int_\Omega \phi\sum_{\beta<\alpha}\binom{\alpha}{\beta}(\text{stuff})\,dx}_{I_2} = \int_\Omega \phi D^\alpha f\,dx.$$

Now subtract I_2 to the other side and you get

$$\begin{aligned}I_1 &= \int_\Omega \phi\left(D^\alpha f-\sum_{\beta<\alpha}\binom{\alpha}{\beta}\left[-\sum_{i,j=1}^n\left(D^\beta u_{x_i}D^{\alpha-\beta}a^{ij}\right)_{x_j}+\sum_{i=1}^n D^\beta u_{x_i}D^{\alpha-\beta}b^i+D^\beta uD^{\alpha-\beta}c\right]\right)\,dx\\ &= \int_\Omega \phi\tilde{f}\,dx = (\tilde{f},\phi).\end{aligned}$$

Now for I_1 we have $\beta = \alpha$ therefore,

$$\begin{aligned} I_1 &= \int_\Omega -\phi \sum_{i,j=1}^n \left(a^{ij} D^\alpha u_{x_i}\right)_{x_j} \, dx + \int_\Omega \sum_{i=1}^n b^i D^\alpha u_{x_i} \phi \, dx + \int_\Omega c D^\alpha u \phi \, dx \\ &= \int_\Omega -\phi \sum_{i,j=1}^n \left(a^{ij} D^\alpha u_{x_i}\right)_{x_j} \, dx + \int_\Omega \sum_{i=1}^n b^i \left(D^\alpha u\right)_{x_i} \phi \, dx + \int_\Omega c D^\alpha u \phi \, dx \\ \text{Integration by Parts on first Integral } &= \int_\Omega \sum_{i,j=1}^n a^{ij} \left(D^\alpha u\right)_{x_i} \phi_{x_j} \, dx + \int_\Omega \sum_{i=1}^n b^i \left(D^\alpha u\right)_{x_i} \phi \, dx + \int_\Omega c D^\alpha u \phi \, dx \\ &= B[D^\alpha u, \phi]. \end{aligned}$$

Therefore, $B[D^\alpha u, \phi] = (\tilde{f}, \phi)$ for each $\phi \in C_0^\infty(\Omega)$, therefore, $D^\alpha u$ is a weak solution to

$$L(D^\alpha u) = \tilde{f}$$

in W, where $D^\alpha u \in H^1(W) = W^{1,2}(W)$. Notice that with (32), (33), (34), and (35) we can get the following for $\tilde{f}$,

$$\begin{aligned} \|\tilde{f}\|_{L^2(W)} &\leq \|D^\alpha f\|_{L^2} + C \sum_{\beta<\alpha} \sum_{i,j=1}^n \|D^\beta u_{x_i}\|_{L^2(W)} + \|D^\beta u_{x_i x_j}\|_{L^2(W)} + C \sum_{\beta<\alpha} \sum_{i=1}^n \|D^\beta u_{x_i}\|_{L^2(W)} + C \sum_{\beta<\alpha} \|D^\beta u\|_{L^2(W)} \\ &\leq \|f\|_{H^{m+1}(\Omega)} + C \sum_{\beta<\alpha} \sum_{i,j=1}^n \|D^\beta u_{x_i}\|_{L^2(W)} + \|D^\beta u_{x_i x_j}\|_{L^2(W)} + C \sum_{\beta<\alpha} \sum_{i=1}^n \|D^\beta u_{x_i}\|_{L^2(W)} + C \sum_{\beta<\alpha} \|D^\beta u\|_{L^2(W)} \\ &\leq \|f\|_{H^{m+1}(\Omega)} + C\|u\|_{H^{m+2}(W)} \\ &\leq \|f\|_{H^{m+1}(\Omega)} + C\|f\|_{H^m(\Omega)} + C\|u\|_{L^2(\Omega)} \qquad (45) \\ &\leq C\left(\|f\|_{H^{m+1}(\Omega)} + \|u\|_{L^2(\Omega)}\right) \qquad (46) \end{aligned}$$

and since $u \in H^1(\Omega)$ and by hypothesis $f \in H^{m+1}(\Omega)$ then $\tilde{f} \in L^2(W)$.

Part 3. Now by the previous Theorem on Interior Regularity we have that $D^\alpha u \in H^2(W) \subset H^2(V)$ and the corresponding estimate coupled with (45) we get

$$\begin{aligned} \|D^\alpha u\|_{H^2(V)} &\leq C\left(\|\tilde{f}\|_{L^2(W)} + \|D^\alpha u\|_{L^2(W)}\right) \\ &\leq C\left(\|f\|_{H^{m+1}(\Omega)} + \|u\|_{L^2(\Omega)}\right). \end{aligned}$$

Notice that the inequality will hold for all $|\alpha| = m+1$; therefore, if $u \in H^{(m+1)+2}(V) = H^{m+3}(V)$ then

$$\|u\|_{H^{m+3}(V)} \leq C\left(\|f\|_{H^{m+1}(\Omega)} + \|u\|_{L^2(\Omega)}\right).$$

□

4 Infinite Differentiability in the Interior

Theorem 4.1 (Infinite Differentiability in the Interior)**.** Assume

$$a^{ij}, b^i, c \in C^\infty(\Omega)$$

for $i, j = 1, \ldots, n$ and $f \in C^\infty(\Omega)$. Suppose $u \in H^1(\Omega)$ is a weak solution of the elliptic PDE $Lu = f$ in Ω. Then $u \in C^\infty(\Omega)$.

Proof. If we assume the hypothesis, that is $a^{ij}, b^i, c, f \in C^\infty(\Omega)$ for all $i, j = 1, \ldots, n$, then obviously $a^{ij}, b^i, c, f \in C^m(\Omega)$ for all $m \in 1, 2, \ldots$. And if we also assume $u \in H^1(\Omega)$ is a weak solution of our elliptic PDE $Lu = f$ in Ω, then by our previous theorem (Theorem 2) we have that $u \in H^{m+2}_{loc}(\Omega)$ for every $m \in 1, 2, \ldots$. Therefore, $u \in H^m_{loc}(\Omega) = W^{m,2}_{loc}(\Omega)$. If we recall Theorem 1.2 and let $p = 2$, then we need to check for two cases: when $m < \frac{n}{2}$ and $m > \frac{n}{2}$.

- If we consider the second case $m > \frac{n}{2}$, then by Theorem 1.2(ii) we have that $u \in C^{m-[n/2]-1,\gamma}(\overline{\Omega})$, but recall that
$$\|u\|_{C^{k,\gamma}(\overline{\Omega})} = \sum_{|\alpha| \leq k} \|D^\alpha u\|_{C(\overline{\Omega})} + \sum_{|\alpha|=k} [D^\alpha u]_{C^{0,\gamma}(\overline{\Omega})}.$$
Therefore, if $u \in C^{m-[n/2]-1,\gamma}(\overline{\Omega})$, then $D^\alpha u \in C(\overline{\Omega})$ for all $|\alpha| < m$, for every $m = 1, 2, \ldots$. Or simply $u \in C^{m-1}(\Omega)$ for all $m = 1, 2, \ldots$, or even more simply $u \in C^\infty(\Omega)$. I left out the closure since we are only dealing with interior regularity.

- If we consider the first case $m < \frac{n}{2}$ and the fact that $u \in W^{m,2}_{loc}(\Omega)$, then $D^\alpha u \in L^2(\Omega)$ for all $|\alpha| \leq m$ for every $m = 1, 2, \ldots,$. Therefore, $D^\alpha u \in C(\Omega)$ for all $|\alpha| < m$, for every $m = 1, 2, \ldots$. And as before $u \in C^\infty(\Omega)$.

□

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